

Unit-3 Corrosion

Q1 What is galvanisation and Tinning?

Ans Both galvanisation and Tinning are metal coating process used to prevent corrosion, but they use different materials & method.

Explain

Galvanisation - The process of Applying a protective zinc coating to iron or steel to prevent rusting.

Tinning - The process of coating a metal (usually copper or iron) with a layer of tin to prevent corrosion & improve solderability.

Q2 Describe Dry / Chemical corrosion?

Ans It occurs due to the direct action of atomic gases like O_2 , CO_2 , H_2S , SO_2 , N_2 and halogens etc. on the surface of metal.

Types

(i) Oxidation

corrosion

metal reacts with O_2 and form metal Oxide.

Ex Rust (Iron Oxide)

(ii) Corrosion by

other gases

metal reacts with sulfur & its compound form sulfides.

Ex $Copper \xrightarrow{S} Copper\ Sulfide$

(iii) Liquid metal

corrosion

metal reacts with halogen (F, Cl)

Ex $metal + Cl_2 \rightarrow metal\ chloride$

Characteristic

- Occurs in the absence of moisture
- High-temperature reaction
- Formation of a protective or non-protective oxide layer
- Common in industrial environment

Prevention

- Use of corrosion resistant materials (stainless steel, aluminum alloys)
- Application of protective coatings (ceramic, oxidation-resistant paint)
- Controlling the environment (reducing temp, using inert gases)

Q3. Write a short note on

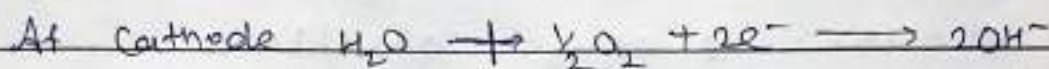
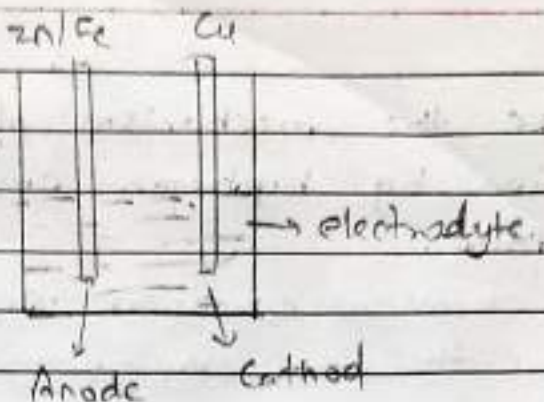
(A) Concentration corrosion

(B) Sacrificial Anodic Protection

Ans (A) Concentration Corrosion:- This type of corrosion takes place due to dissimilarity of a solution, where a solution may involve high concentration of oxygen at one point of the metal & low at other.

→ Where the Conc. of O_2 is high, e^- are released and it becomes anode.

→ e^- flows towards low conc. of O_2 & it becomes cathode. Thus the medium becomes electrolyte.

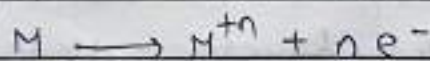


overall reaction



- (B) Sacrificial Anodic Protection: This is a type of electrochemical process used in corrosion protection. In this method, a more reactive metal (sacrificial Anode) is intentionally allowed to corrode instead of a protected metal structure (steel pipelines / ship hulls etc)

reaction -



M = sacrificial metal

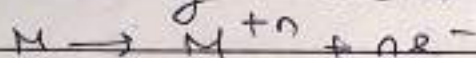
e^- released travel to the protected metal, reducing the risk of its oxidation corrosion.

Q4 Describe the mechanism of electrochemical corrosion by hydrogen evolution to oxygen absorption.

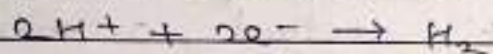
Ans Electrochemical corrosion occurs when a metal reacts with its environment through electrochemical processes, leading to its gradual deterioration. The two common mechanisms of evolution & oxygen absorption.

iv Hydrogen Evolution :- (Acidic Medium)

- Anodic reaction - metal dissolves, releasing electrons.



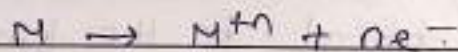
- Cathodic reaction: Hydrogen ions (H^{+}) gain electrons to form hydrogen gas.



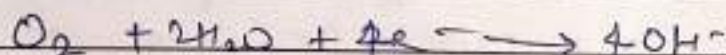
- No oxide formation, metal dissolves directly.

(ii) Oxygen Absorption :- (Neutral / Alkaline Medium)

- Anodic Reaction - Metal dissolves, releasing electrons.



- Cathodic Reaction - O_2 reduces to hydroxide ions.



- Forms corrosion products like rust ($Fe_2O_3 \cdot 2H_2O$ for iron)

Q5 What is Pitting. Bedworth rule?

Ans Pitting bed worth rule suggests that metals with an unfavorable PBR are more likely to suffer from pitting corrosion, which is a highly localized and ~~and~~ dangerous form of deterioration.

Pitting Bedworth ratio (PBR)

$$PBR = \frac{\text{Volume of oxide}}{\text{Volume of metal consumed.}}$$

if $PBR < 1 \Rightarrow$ the oxide layer is thin & non-protective

$PBR > 1 \Rightarrow$ the oxide layer is thick & too

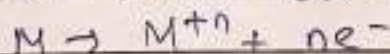
brittle.

Q6 Explain anodic sacrificial protection to minimize the corrosion.

Ans Anodic sacrificial protection is a cathodic protection method used to prevent corrosion in metals especially iron & steel structures, by using a more reactive metal (called a sacrificial anode). The sacrificial anode oxidizes (corrodes) preferentially protecting the main structure from corrosion.

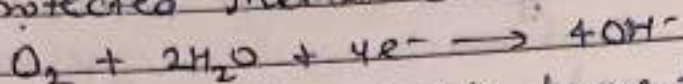
In this method more electrochemically active metal is chosen (higher tendency to oxidize). Common sacrificial anode materials zinc (Zn), magnesium (Mg), Aluminium (Al).

At the Sacrificial Anode (Oxidation)



The anode metal loses electrons & corrodes.

At the protected metal (Cathode - Reaction):



(Electrons from the anode prevent oxidation of the main metal.)

Electron flow and to the protected metal, this makes the protected metal act as a cathode, preventing it from corroding.

This is cost effective & simple method but require periodic replacement of the sacrificial anodes as they corrode over time.

Q.7 What is the reason of corrosion? Explain the effects of corrosion consequence?

Ans Corrosion is the deterioration of metal due to their reaction with the environment typically involving oxidation. It leads to the formation of rust, tarnish or other degradation products weakening the metal structure over time.

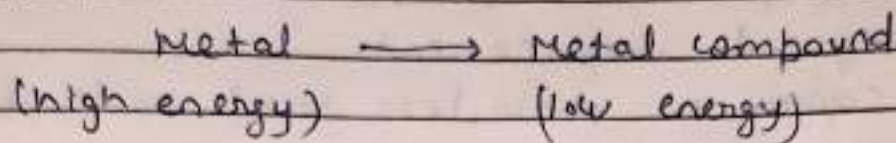
Causes of Corrosion -

- (i) Electrochemical Reaction - metal react with O_2 and moisture, forming rust.
- (ii) Environmental factor - Humidity, temperature, pollution, accelerate corrosion.
- (iii) Galvanic Corrosion - Contact between different metal in an electrolyte speed up

Corrosion:

- (iv) Microbial Corrosion - Bacteria produce acids that damage metals.

→ Metal are in higher energy state. So they have tendency to convert into lower energy state.



Effect of Corrosion:-

- (i) Structural damage - bridges, pipelines, building weaken.
- (ii) Economic Loss - High maintainance & repair cost.
- (iii) Safety Hazards - leaks, explosions, equipment failure
- (iv) Reduced efficiency - Machinery performance decrease.
- Environmental impact - Pollution from corroded material.

Q.8 Copper equipment should not possess steel bolt? give a reason.

Ans Because of galvanic corrosion, Copper equipment shouldn't possess steel bolt.

When two different metals (copper + steel) are in contact in the presence of an electrolyte (moisture, water, air humidity), a galvanic cell forms. In this setup, steel is less noble than copper & acts as the anode, while copper acts as the cathode. The anodic metal (Steel bolt) corrodes rapidly, leading to failure.

reaction



Date: / /

Steel bolts weaken & fail, reducing structural integrity, Corrosion spreads, affecting the copper equipment, Increased maintenance costs due to frequent replacement.
So copper or stainless steel bolts to avoid galvanic corrosion.